Pass-Through

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Fall 2026

Grad IO

What is pass-through and why do we care?

Pass-Through: What is it?

Lots of cases in economics where we want to know how prices respond to changes in costs:

- Response to cost shocks (oil shock, commodity prices)
- Response to tax changes (incidence/efficiency of taxes)
- Transmission of exchange rate shocks
- Transmission of monetary policy
- Price effects of mergers: converting UPP (a change in opportunity cost) into a change in price
- Double marginalization: how much of a wholesale price change reaches the shelf

Today: what determines pass-through in theory, what the demand models of this course imply for it, and why it is so hard to estimate.

Hot take

Pass-through is the simplest thing in economics that nobody understands

- Often we are talking about different things
  - IO: mostly levels, $\rho = \frac{\partial p_j}{\partial mc_j}$ or the matrix $\frac{\partial \mathbf{p}}{\partial \mathbf{mc}}$
  - Macro/Trade: mostly logs, $\rho^{\log} = \frac{\partial \log p_j}{\partial \log mc_j}$
  - These are not the same number. With $\mu_j \equiv p_j/mc_j$: $\rho_j^{\log} = \rho_j \cdot \frac{mc_j}{p_j} = \frac{\rho_j}{\mu_j}$.
    Complete pass-through in levels is incomplete pass-through in logs whenever $p > mc$ (Sangani 2026).
  - (Throughout this lecture ρ is the pass-through rate – not the nesting parameter from the demand lectures.)
- One thing we know for sure: constant marginal costs, perfect competition:

$$p = mc \rightarrow \frac{\partial p}{\partial mc} = 1$$

- Everything else depends on two things:
  - Curvature of demand (second derivatives)
  - Nature of competition (conduct)
- Neither is something an own-price elasticity pins down. That is the whole lecture.

The Empirical Puzzle: Estimates Are All Over the Place

Shock → price	Units	$\hat{\rho}$	Source
Exchange rate → US import prices, dollar-priced	logs	0.25	Gopinath Itskhoki Rigobon 2010
same, priced in foreign currency	logs	0.95	
Exchange rate → retail beer	logs	≈ 0.25	Goldberg Hellerstein 2013
Commodity (coffee) → retail	logs	≈ 0.30	Nakamura Zerom 2010
Cigarette taxes	levels	< 1 / ≈ 1	Harding et al. 2012 / DeCicca et al. 2013
Gasoline taxes	levels	≈ 1; holidays $\ll 1$	Marion Muehlegger 2011; Doyle Samphantharak 2008
Sales taxes	ad val.	≈ 1, half > 1	Poterba 1996; Besley Rosen 1999
Alcohol taxes	levels	1.5 – 2.1	Young Bielinska-Kwapisz 2002; Kenkel 2005

- Trade theory (CES + monopolistic competition) predicts 0 or 1 depending on invoicing currency. The data say 0.25 and 0.95.
- Read the units column: the low numbers are in logs, the numbers near one are in levels. Nakamura–Zerom's 0.30 for coffee is one-for-one in cents (next slides).
- Anything from 0.25 to 2 shows up. Theory says that is exactly what to expect: pass-through is about curvature and conduct, and both vary across markets.
- Many "price elasticity" studies (gasoline, cigarettes, alcohol) assume $\rho = 1$ and regress quantities on tax changes: $\frac{\partial \log q}{\partial \log t} = \frac{\partial \log q}{\partial \log p}$ (!)

Logs vs. Levels: Sangani (QJE 2026)

Leontief, constant returns: unit cost $c + w$ (commodity c, everything else w). Two pricing rules:

percentage markup: $p = \mu\,(c + w)$ additive markup: $p = c + w + m$

	Initial	Change	New	% Change	$\rho^{\log}$ (wrt mc)	ρ (levels)
Commodity cost c	\$1	$+\$0.20$	\$1.20	$+20$		
Marginal cost $c + w$	\$2	$+\$0.20$	\$2.20	$+10$		
Price, percentage markup ($\mu = 2$)	\$4	$+\$0.40$	\$4.40	$+10$	1.0	2.0
Price, additive markup ($m = \$2$)	\$4	$+\$0.20$	\$4.20	$+5$	0.5	1.0

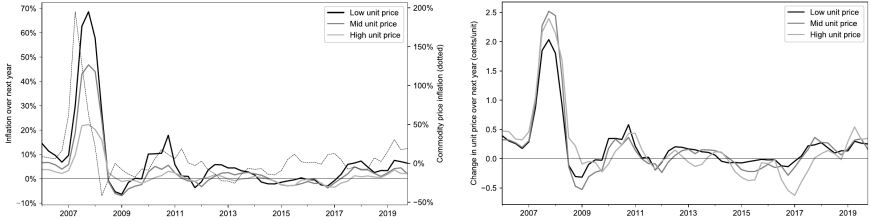
- Regressing $\Delta \log p$ on $\Delta \log c$ estimates $\rho \cdot \frac{c}{p}$: level pass-through times the commodity's share of the price (not of cost). A fixed percentage markup means $\rho = \mu > 1$ in levels.
- Log pass-through is not a stable statistic when markups are additive. For a firm whose whole marginal cost c moves, with $dp = dc$ and $\mu = p/c$, to second order (his Prop. 1):

$$\rho^{\log} = \frac{\Delta \log p}{\Delta \log c} \approx \frac{1}{\mu}\left[1 + \frac{1}{2}\frac{\mu - 1}{\mu}\,\Delta \log c\right]$$

  so a log regression manufactures asymmetry (increases $>$ decreases: "rockets and feathers"), size dependence (big shocks $>$ small ones), and lower pass-through for high-markup firms (large firms, high quality), with no change in behavior.
- Scope: common (industry-wide) shocks to relative input prices. Idiosyncratic shocks are a different experiment (later).

Sangani (2026): Complete Pass-Through in Levels, Incomplete in Logs

- Perth gasoline (weekly, station level, 2001–2022): long-run pass-through 0.991 (0.038) in levels; 0.899 (0.043) in logs, below gasoline's cost share (≥ 0.97).
- Interact with station markups: in levels the interaction is ≈ 0; in logs it is -0.010 to -0.015 per point of net markup (OLS and IV), what additive markups predict (≈ -0.01). As markups $\to 0$, log pass-through $\to$ the cost share.
- Food (BLS average prices, 12 months), logs / levels: coffee 0.47 / 0.95, rice 0.31 / 0.88, flour 0.24 / 0.87, OJ 0.33 / 0.97, beef 0.41 / 0.90; sugar 0.37 / 0.69 is the exception.
- Manufacturing (NBER-CES, 459 industries): $\Delta \log p$ on $\Delta \log c \times \frac{\text{input cost}}{\text{sales}}$ gives $\rho \approx 0.95$–1.04 (eq. (9) of the paper).



(A) Inflation (in percentages)

(B) Price change (in levels)

Rice in NielsenIQ, by tercile of unit price. 2008 run-up: cheap rice inflates 70%, expensive rice under 25% (left); all three rise by the same cents per unit (right).

One product, one price

Perfect Competition: Jenkin (1872), Alfred Marshall (1890)

$$D(p) = S(p - t)$$

$$\rho = \frac{dp}{dt} = \frac{1}{1 + \frac{\varepsilon_d}{\varepsilon_s}}$$

- Where ε_d is demand elasticity and ε_s is supply elasticity respectively.
- $\rho \in [0, 1]$ as long demand slopes down and supply slopes up!
- Constant MC implies that $\varepsilon_s \to \infty$ and $\rho \to 1$
- Incidence $I = \frac{\frac{dCS}{dt}}{\frac{dPS}{dt}} = \frac{\rho}{1-\rho}$ (higher pass-through, more borne by consumers)

So far so good. Nothing about curvature yet: with perfect competition only slopes matter.

Monopoly (Weyl and Fabinger, JPE 2013)

Start with $MR = MC$ and differentiate with respect to the tax t:

$$p(q) + p'(q)\,q = mc(q) + t \quad \Rightarrow \quad \frac{dq}{dt} = \frac{1}{mr' - mc'} \quad \Rightarrow \quad \rho = \frac{dp}{dt} = p'\frac{dq}{dt} = \frac{p'}{mr' - mc'}$$

Define the markup term $ms(q) \equiv -p'(q)\,q$ and the elasticities $\epsilon_D = -\frac{p}{p'q}$, $\epsilon_S = \frac{mc}{mc'q}$, $\epsilon_{ms} = \frac{ms}{ms'q}$. Using the Lerner index $\frac{mc}{p} = \frac{\epsilon_D - 1}{\epsilon_D}$:

$$\boxed{\rho = \frac{1}{1 + \frac{\epsilon_D - 1}{\epsilon_S} + \frac{1}{\epsilon_{ms}}}} \qquad \text{constant MC: } \rho = \frac{1}{1 + 1/\epsilon_{ms}}, \qquad \frac{1}{\epsilon_{ms}} = 1 + \frac{p''q}{p'}$$

- $\epsilon_D > 1$ for a monopolist (except for my MBA students).
- $1/\epsilon_{ms}$ is the curvature of (inverse) demand, and $(\log D)'' = -\frac{1}{\epsilon_{ms}}\frac{1}{ms^2}$, so its sign is the sign of the curvature of log demand:
  - log-concave demand $\Rightarrow 1/\epsilon_{ms} > 0 \Rightarrow \rho < 1$ (linear demand: $\rho = 1/2$)
  - log-linear (exponential) demand $\Rightarrow \rho = 1$
  - log-convex demand $\Rightarrow \rho > 1$ (constant elasticity)
- Bulow and Pfleiderer (1983): with constant MC a monopolist's pass-through is entirely about curvature.

Canonical Demand Curves (single product, constant MC)

Demand	$\rho = dp/dc$	log D	Markup
Perfect competition	1	–	0
Linear $q = a - bp$	$1/2$	concave	$(p - c) = \frac{a-bc}{2b}$
Exponential $q = Ae^{-\lambda p}$	1	linear	$p - c = 1/\lambda$ (constant)
Constant elasticity $q = Ap^{-\epsilon}$	$\frac{\epsilon}{\epsilon-1} > 1$	convex	$p = \frac{\epsilon}{\epsilon-1} c$ (constant Lerner)
Logit, single-product firm, rivals fixed	$1 - \sigma_j$	concave	$p_j - c_j = \frac{1}{\alpha(1-\sigma_j)}$

- Linear, exponential and constant-elasticity are the $\rho = \frac{1}{2}, 1, > 1$ members of the Bulow–Pfleiderer (1983) constant pass-through family; the puzzle slide lives between the linear and CES rows.
- A constant absolute markup (exponential) gives $\rho = 1$; a constant percentage markup (CES) gives $\rho > 1$ because the markup scales with cost. Hence CES trade models over-shift unless pricing-to-market or invoicing is added. These are the two benchmarks on the logs-vs-levels slide: complete in levels vs. complete in logs.
- Logit sits below one and gets closer to one as the product gets smaller. Derivation in two slides.

Conduct and Incidence (Weyl and Fabinger 2013)

Generalize to symmetric imperfect competition with a conduct parameter θ: $\theta = 0$ (PC), $\theta = 1$ (monopoly), $\theta = 1/N$ (symmetric Cournot):

$$\rho = \frac{1}{1 + \frac{\theta}{\epsilon_\theta} + \frac{\epsilon_D - \theta}{\epsilon_S} + \frac{\theta}{\epsilon_{ms}}} \qquad \text{with } \epsilon_\theta \equiv \frac{\theta}{\theta' q} \text{ (how conduct itself changes with quantity).}$$

Incidence follows from pass-through and conduct alone (their "Principle of Incidence"):

$$\frac{1}{I} = \frac{1}{\rho} - (1 - \theta) \iff I = \frac{\rho}{1 - (1 - \theta)\rho} : \qquad I_{PC} = \frac{\rho}{1 - \rho}, \qquad I_{\text{monopoly}} = \rho.$$

- Monopoly: producer surplus falls by exactly q per dollar of tax (envelope theorem), consumer surplus by $\rho\, q$. So $\rho > 1$ means consumers bear more than the whole tax.
- Two sufficient statistics for incidence, (ρ, θ), neither of which is an elasticity.
- Not sure how I feel about the revival of the conduct parameter (more in the conduct lecture).

Elasticity vs. Curvature: Mrázová and Neary (AER 2017)

Write everything in terms of inverse demand $p(x)$. Two local statistics at any point:

$$\varepsilon(x) \equiv -\frac{p(x)}{x\,p'(x)} > 0 \quad \text{(elasticity)} \qquad \kappa(x) \equiv -\frac{x\,p''(x)}{p'(x)} \quad \text{(convexity; MN call it } \rho\text{)}$$

Monopoly, constant MC: $\dfrac{dp}{dc} = \dfrac{1}{2-\kappa} \quad \Rightarrow \quad \dfrac{dp}{dc} - 1 = \dfrac{\kappa - 1}{2-\kappa}$

- Second-order condition: $\kappa < 2$. Over-shifting ($\rho > 1$) iff $\kappa > 1$: demand more convex than exponential.
- Superconvex demand ($\kappa > 1 + 1/\varepsilon$, more convex than CES at the same elasticity): the proportional markup p/c *rises* with cost. CES is the boundary.
- The demand manifold: any demand function traces a curve in (κ, ε) space. The elasticity sets the markup, the convexity sets the pass-through, and knowing one tells you nothing about the other.
- Our problem in one line: demand estimation targets ε; cost-change counterfactuals need κ, which the functional form supplies for free.

Mrázová and Neary (2017): The Space of Elasticity and Convexity

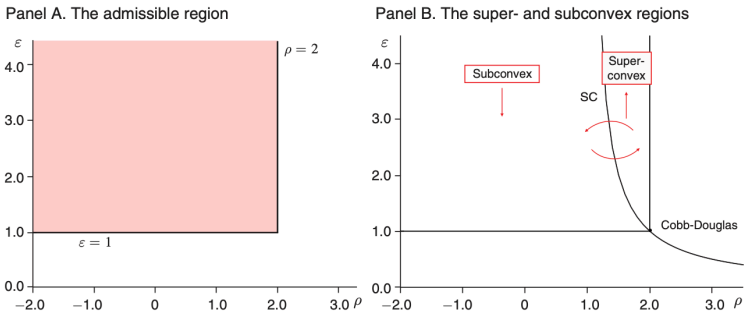


FIGURE 1. THE SPACE OF ELASTICITY AND CONVEXITY

Panel A: the admissible region ($\varepsilon > 1$, $\kappa < 2$). Panel B: the SC curve is the CES family (ε constant); below it demand is subconvex (proportional markups fall with cost), above it superconvex. Cobb–Douglas is the corner.

Discrete choice

Logit Pass-Through in Two Lines

Single-product firm, plain logit, rivals' prices held fixed. The FOC is $p_j = c_j + \eta_j(p_j)$ with

$$\eta_j = \frac{1}{\alpha(1-\sigma_j)}, \qquad \frac{\partial \eta_j}{\partial p_j} = \frac{1}{\alpha}\frac{1}{(1-\sigma_j)^2}\frac{\partial \sigma_j}{\partial p_j} = -\frac{\sigma_j}{1-\sigma_j} \qquad \Rightarrow \qquad \rho_j = \frac{1}{1-\partial \eta_j/\partial p_j} = \boxed{1-\sigma_j}$$

- Logit demand is log-concave in own price: $(\log \sigma_j)'' = -\alpha^2 \sigma_j(1-\sigma_j) < 0$, so $\rho_j < 1$ always.
- The markup $\frac{1}{\alpha(1-\sigma_j)}$ *falls* as the price rises (share shrinks), which is what absorbs part of the cost increase. A niche product passes through almost everything; a dominant product very little.
- Note what is not in the formula: α. The price coefficient sets the level of markups and elasticities; the curvature, and hence pass-through, is pinned down by the share alone. This is the second-derivative analogue of IIA.
- With rivals responding (strategic complements) own pass-through is higher and cross pass-through is positive: that needs the matrix (next section). The $1-\sigma_j$ is its "rivals fixed" diagonal. A cost shock common to every product is a third experiment, with its own answer (after the matrix).

Mixed Logit: Curvature Comes From the Mixing Distribution

Aggregate demand is a mixture, $\sigma_j(p) = \sum_i w_i \, \sigma_{ij}(p)$, and mixtures of log-concave functions need not be log-concave.

- Raise p_j: the price-sensitive types (high α_i) leave first, so the residual demand is less elastic than the average. That selection makes $\log \sigma_j$ more convex than any single type's, and $\rho_j > 1$ becomes possible.
- So in a BLP model pass-through is a byproduct of the distribution you assumed for α_i (and the characteristics with random coefficients), which estimation on prices and shares barely disciplines. Normal, lognormal, or a two-point α_i can all fit the same elasticities and imply different pass-through.
- Miravete, Seim, Thurk (2023, "Elasticity and Curvature of Discrete Choice Demand Models"): map the (ε, κ) pairs each specification can reach in equilibrium. Plain logit and nested logit are a curve; mixed logit is a region whose shape is set by the mixing family, not the data.
- Practical advice: (a) report the implied pass-through next to the elasticities, it is a cheap diagnostic; (b) if you have cost or tax variation, use it as a supply moment (Conlon Rao in the BLP lecture matched observed markups and a tax change for exactly this reason); (c) if your counterfactual is a cost change, the mixing distribution is doing the work, so vary it.

Many products: the pass-through matrix

Multiproduct Bertrand: Pass-Through Is a Matrix

Recall the FOC in markup form from the merger simulation lecture, $\Omega(\mathbf{p}) \equiv \mathcal{H} \odot \Delta(\mathbf{p})$:

$$f(\mathbf{p}, \mathbf{mc}) \equiv \mathbf{p} - \mathbf{mc} - \underbrace{\Omega(\mathbf{p})^{-1}\boldsymbol{\sigma}(\mathbf{p})}_{\boldsymbol{\eta}(\mathbf{p})} = 0.$$

The implicit function theorem, with $\partial f / \partial \mathbf{mc} = -\mathbb{I}_J$ because costs enter additively:

$$\mathcal{P} \equiv \frac{\partial \mathbf{p}}{\partial \mathbf{mc}} = -\left(\frac{\partial f}{\partial \mathbf{p}}\right)^{-1} \frac{\partial f}{\partial \mathbf{mc}} = \left(\mathbb{I}_J - \frac{\partial \boldsymbol{\eta}}{\partial \mathbf{p}}\right)^{-1}, \qquad \mathcal{P}_{(j,k)} = \frac{\partial p_j}{\partial mc_k}.$$

- Diagonal: own-cost pass-through, now including rivals' equilibrium responses. Off-diagonal: a rival's cost shock moves my price (positive under strategic complements).
- Single-product logit: $\partial \eta_j / \partial p_j = -\sigma_j/(1-\sigma_j)$ on the diagonal; ignoring the off-diagonal terms gives back $1 - \sigma_j$.
- Multiproduct firms: η_j depends on the margins of all sister products, so a cost shock to k moves p_j with no rival response at all.
- Everything hard is in $\partial \boldsymbol{\eta} / \partial \mathbf{p}$: the derivative of a markup is a second derivative of demand.

How do we get the pass-through matrix?

Two ways:

1. Numerically: perturb the marginal cost of each product k in turn and re-solve for equilibrium prices

$$\mathbf{p} = \mathbf{mc} + e_k + \Omega(\mathbf{p})^{-1}\boldsymbol{\sigma}(\mathbf{p})$$

   (Morrow Skerlos (2011) fixed point, J solves, finite differences; fine for small J.)

2. Analytic: differentiate the FOC and apply the implicit function theorem (next slide).
   See Jaffe Weyl (AEJM 2013), Miller Weinberg (2017, Appendix E), Conlon Rao (2025).
   This is what PyBLP does with `results.compute_passthrough`, building the full $J \times J \times J$ demand Hessian (slow, and J^3 memory). JaxBLP contracts that tensor with the markups analytically first and never forms it ($J^2 I$ flops; 0.4s vs. 3.7s and 31GB on a 636-product market).

Multivariate IFT: The Hard Part

Differentiate the FOC with respect to p_l to get column l of $\partial f / \partial \mathbf{p}$:

$$\frac{\partial f(\mathbf{p}, \mathbf{mc})}{\partial p_l} \equiv e_l + \Omega^{-1}(\mathbf{p}) \left[\mathcal{H} \odot \frac{\partial \Delta(\mathbf{p})}{\partial p_l} \right] \Omega^{-1}(\mathbf{p}) \, \boldsymbol{\sigma}(\mathbf{p}) - \Omega^{-1}(\mathbf{p}) \, \frac{\partial \boldsymbol{\sigma}(\mathbf{p})}{\partial p_l}. \tag{1}$$

The complicated piece is the demand Hessian: a $J \times J \times J$ tensor with elements (j, k, l), $\frac{\partial^2 \sigma_j}{\partial p_k \partial p_l}$, since $\frac{\partial \Delta(\mathbf{p})}{\partial p_l} = -\frac{\partial^2 \boldsymbol{\sigma}}{\partial \mathbf{p} \partial p_l}$.

- Compactly, with markups $\boldsymbol{\eta} = \Omega^{-1} \boldsymbol{\sigma}$: $\frac{\partial f}{\partial \mathbf{p}} = \mathbb{I}_J + \Omega^{-1} \left[T - \frac{\partial \boldsymbol{\sigma}}{\partial \mathbf{p}} \right]$ with $T_{(j,l)} = \sum_m \frac{\partial \Omega_{(j,m)}}{\partial p_l} \eta_m$. The tensor only ever appears contracted with the markups; computing T directly is what makes the closed form cheap.
- Orientation (as in the pricing lecture): with product-specific α_{ik}, read $\Omega = \mathcal{H} \odot \Delta'$.
- For (mixed) logit the Hessian is a sum of products of σ_{ij}'s and α_i's: the "curvature from mixing" slide, in matrix form.

Application: From UPP to Price Effects (Jaffe Weyl 2013)

Recall UPP from the antitrust deck: $UPP_j = \Delta c_j + \sum_{k \in \mathcal{J}_g} (p_k - c_k) \cdot D_{jk}(\mathbf{p})$ is a change in opportunity cost. But how much do we expect prices to rise?

- Treat the merger as a cost shock and pass it through: $\Delta \mathbf{p} \approx \mathcal{P} \cdot \mathbf{UPP}$. The first-order approach to merger analysis.
- That needs the whole $J \times J$ matrix $\mathcal{P}$, which is J^2 numbers (again). So people restrict it: diagonal, or $\mathcal{P} = 0.8 \cdot \mathbb{I}_J$, or just the merging firms' own-cost pass-through.
- Miller, Remer, Ryan, Sheu (2016) simulate mergers under several demand systems and ask how good the approximation is with the full matrix, the diagonal, or only the merging firms' rows (next slide).
- Which approximations work depends on demand curvature, and curvature is exactly what the approximation is trying to avoid estimating.

Miller, Remer, Ryan, Sheu (2016): How Good Is the First-Order Approximation?

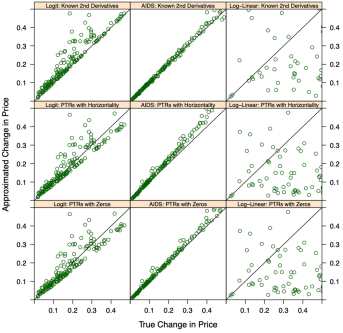


Figure 2: Prediction Error with Complete Information.

Notes: The figure provides scatter-plots of approximation against the true price effect for logit demand, the AIDS and log-linear demand. The case of linear demand is omitted because approximation is exact in that setting. Approximations are calculated alternately based on full knowledge of the second-order properties of demand in neighborhood of pre-merger equilibrium ("Known 2nd Derivatives"); based on full knowledge of pre-merger cost pass-through and the horizontality assumption ("PTRs with Horizontality"); and based on full knowledge of pre-merger cost pass-through and the assumption that derivatives of the form $\partial^2 Q_m/\partial P_l \partial P_k$ equal zero ("PTRs with Zeros").

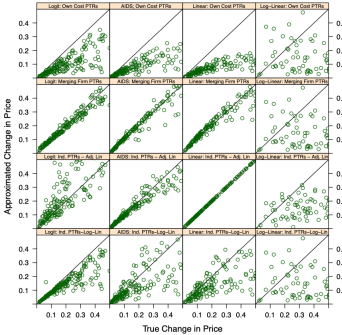


Figure 3: Prediction Error with Incomplete Information

Notes: The figure provides scatter-plots of approximations against the true price effect for logit demand, the AIDS, linear demand and log-linear demand. Four informational scenarios are considered: pre-merger cost pass-through that is available only for own costs ("Own Cost PTRs"); pre-merger cost pass-through that is available only for the merging firms ("Merging Firms' PTRs"); industry cost pass-through that is apportioned using the adjusted-linear method ("Ind. PTRs–Adj.–Lin."); and industry cost pass-through that is apportioned using the log-linear method ("Ind. PTRs–Log–Lin").

Left (complete information): with the full pass-through matrix the approximation is close to exact for AIDS and acceptable for logit; log-linear (CES-like) demand is hopeless because the price effect is far from linear in the cost shock. Right (incomplete information): using only own-cost pass-through understates every price effect; the merging firms' rows do most of the work.

Common shocks: shift vs. scale invariance

Which Demand Systems Pass a Common Shock Through in Levels? (Sangani 2026)

A common shock raises every inside good's cost by dc (outside prices fixed). Its price response is the row sum $\mathcal{P}\mathbf{1}$, not the diagonal.

- Scale invariance: $D_j(\lambda\mathbf{p}) = \lambda^{\varphi_j} D_j(\mathbf{p})$ with φ_j not depending on p_j. A proportional shift leaves every elasticity unchanged $\Rightarrow$ complete in logs, so $\rho = \mu > 1$ in levels (Prop. 4).
  All homothetic demand: CES, Kimball, HSA, nested CES (Atkeson–Burstein), nested Kimball (Amiti–Itskhoki–Konings). Their variable markups only show up for idiosyncratic shocks.
- Shift invariance: $D_j(\mathbf{p} + \lambda\mathbf{1}) = (1 + \lambda\psi_j)\, D_j(\mathbf{p})$ with ψ_j not depending on p_j. A uniform shift rescales each residual demand curve, leaving the additive markup $D_j/|\partial D_j/\partial p_j|$ unchanged $\Rightarrow$ complete in levels (Prop. 5). No demand system is both (Prop. 6).
- Leading case, $\psi_j = 0$: quasi-linear random utility $u_{ij} = \delta_{ij} - p_j$ where every consumer buys one of the inside goods (McFadden's translation invariance; Anderson, de Palma, Thisse 1992). That covers logit, nested logit, mixed logit with any mixing distribution, probit, Hotelling, and Salop.

In our notation: if $\boldsymbol{\sigma}(\mathbf{p} + \lambda\mathbf{1}) = \boldsymbol{\sigma}(\mathbf{p})$, then Δ and $\boldsymbol{\eta} = \Omega^{-1}\boldsymbol{\sigma}$ depend only on price *differences*, so

$$\frac{\partial \boldsymbol{\eta}}{\partial \mathbf{p}}\mathbf{1} = \mathbf{0} \quad \Rightarrow \quad \left(\mathbb{I}_J - \frac{\partial \boldsymbol{\eta}}{\partial \mathbf{p}}\right)\mathbf{1} = \mathbf{1} \quad \Rightarrow \quad \boxed{\mathcal{P}\,\mathbf{1} = \mathbf{1}}$$

for any ownership matrix $\mathcal{H}$. The diagonal of $\mathcal{P}$ can be anything the curvature says; the row sums cannot.

The Outside Good Is What Breaks It

An outside good at a fixed price breaks translation invariance: a common increase pushes consumers out of the market. (So do income effects, $\alpha \log(y_i - p_j)$ as in BLP 1995 and Petrin 2002.) Take symmetric single-product logit with inside shares σ_j and outside share σ_0. Raise all J prices by dp:

$$d\sigma_j = -\alpha\, \sigma_j \sigma_0\, dp, \qquad d\eta_j = d\left[\frac{1}{\alpha(1-\sigma_j)}\right] = -\frac{\sigma_j \sigma_0}{(1-\sigma_j)^2}\, dp$$

$$\Rightarrow \quad \rho^{\text{common}} = \frac{1}{1 + \frac{\sigma_j}{1-\sigma_j}\, D_{j0}} \quad \text{vs.} \quad 1 - \sigma_j = \frac{1}{1 + \frac{\sigma_j}{1-\sigma_j}}$$

where $D_{j0} = \sigma_0/(1-\sigma_j)$ is logit diversion to the outside good.

- No diversion to the outside good $\Rightarrow$ complete in levels. Atomistic firms ($\sigma_j \to 0$) also give complete pass-through in levels: $p_j = c_j + p_0/b_j$ in Sangani's logit.
- Mixed logit with an outside good can give anything. Numerical example: 4 products, three types $\alpha_i \in \{0.5, 1.5, 4\}$, $\sigma_0 \approx 0.72$. Row sums of $\mathcal{P}$ are 1.76–1.89, because the price-sensitive types leave first. Drop the outside good, keep the same (δ, α_i): exactly 1.
- Own and common pass-through use different curvature: residual-demand superelasticity vs. that plus how rivals' prices move j's elasticity. (Nakamura–Zerom's 4.64, in Kimball demand, still gives complete *log* common pass-through.)
- For estimation: level pass-through of industry-wide shocks is a cheap moment that disciplines what prices and shares do not, the outside-good margin and the mixing distribution.

Why Logs vs. Levels Matters Beyond Measurement (Sangani 2026)

- Margins and entry. Percentage markups plus inelastic industry demand mean a cost increase raises profit per unit, so operating margins or entry must rise. Data: gross margins fall with input prices (elasticity -0.26 to -0.29 for retail gasoline, -0.19 for NBER-CES manufacturing). Operating margins and establishment counts do not move. Additive markups predict exactly this.
- Dispersion. Additive markups mean a common cost increase compresses the dispersion of log prices, percentage markups, and TFPR / labor-productivity dispersion, with *no* reallocation of quantities (rice, flour, coffee). Under non-homothetic demand, falling markup dispersion is not evidence of less misallocation.
- Measured markups. Percentage markups and Lerner indices fall mechanically when input prices rise, with no change in conduct. Keep this in mind when reading markup time series.
- Macro. Input–output model of the US (402 industries, Calvo). With CES (percentage) markups, CPI inflation has a standard deviation of 2.5% vs. 1.1–1.3% in the data. Calling labor a variable cost lowers it, but implies markups of 1.17–1.31. Logit (additive) markups give 1.3% and match Känzig's oil-shock response of the CPI (peak 0.43 vs. 0.45).
- Okun (1981): firms pass materials through "on a dollars-and-cents basis" and labor "with a percentage markup." In logit with α measured against the wage, the additive markup is w/b, so wage shocks move it.

Vertical chains

Double Marginalization: Villas-Boas (ReStud 2007), Miller Weinberg (Ecma 2017)

Most retail prices are set by a retailer who buys from a manufacturer with market power. Retailer and wholesaler FOCs:

$$\mathbf{p^r} = \underbrace{\mathbf{p^w} + \mathbf{c^r}}_{\mathbf{mc^r}} + (\mathcal{H}_r \odot \Delta_r(\mathbf{p^r}))^{-1}\boldsymbol{\sigma}(\mathbf{p^r})$$

$$\mathbf{p^w} = \mathbf{mc^w} + \left(\mathcal{H}_w \odot \left(\left[\frac{\partial \mathbf{p^r}}{\partial \mathbf{p^w}}\right]' \cdot \Delta_r(\mathbf{p^r})\right)\right)^{-1} \boldsymbol{\sigma}(\mathbf{p^r})$$

- Δ_r is the matrix of (retail) demand derivatives with entries $-\frac{\partial \sigma_j}{\partial p_k^r}$ (as in the merger simulation lecture).
- $\mathcal{H}_r, \mathcal{H}_w$ ownership matrix $(j,k) = 1$ if both products sold by same retailer/wholesaler.
- $\frac{\partial \mathbf{p^r}}{\partial \mathbf{p^w}}$ is the pass-through matrix of the retail stage, entry $(m,j) = \partial p_m^r / \partial p_j^w$. Chain rule: $-\partial\sigma_k/\partial p_j^w = \sum_m \frac{\partial p_m^r}{\partial p_j^w}\Delta_{r,(m,k)}$, so the wholesaler's matrix is the transpose of the pass-through times Δ_r (easy to get wrong in code).

Challenge: We want $\mathbf{p^r}(\mathbf{p^w})$ and $\mathbf{mc^w}$ but we only have an implicit solution for the retailer FOC. The pass-through matrix is the object that lets the upstream firm's problem be written at all.

Pass-through Counterfactuals? (Conlon Rao 2025)

How do we solve for $\mathbf{p^w}$ under a counterfactual pass-through matrix?

- Idea: pass-through only augments the matrix $\Delta_r(\mathbf{p^r})$: $\Omega_w \equiv \mathcal{H}_w \odot (\mathcal{P}'\Delta_r)$ with $\mathcal{P} \equiv \frac{\partial \mathbf{p^r}}{\partial \mathbf{p^w}}$.
- Example: a constant retail markup or sales tax, $\mathcal{P} = \text{diag}(1+\tau_r)$, so $\mathcal{P}' = \mathcal{P}$ and $\mathcal{P}'\Delta_r = \mathcal{P}\Lambda - \mathcal{P}\Gamma$ with $\mathcal{P}\Lambda$ still diagonal.

$$\mathbf{p^w} = \mathbf{mc^w} + \Omega_w^{-1}\boldsymbol{\sigma}(\mathbf{p^r})$$

Adapt the Morrow Skerlos ζ fixed point (same sign convention as the pricing lecture, $\alpha_i > 0$). For diagonal $\mathcal{P}$, $\mathcal{H}_w \odot (\mathcal{P}\Gamma) = \mathcal{P}(\mathcal{H}_w \odot \Gamma)$, so $\mathcal{P}$ cancels out of the Γ term and survives only on the share term:

$$\mathbf{p^w} \leftrightarrow \mathbf{mc^w} + \boldsymbol{\zeta}(\mathbf{p^w}) \quad \text{where} \quad \boldsymbol{\zeta}(\mathbf{p^w}) = \Lambda^{-1}\left[\mathcal{H}_w \odot \Gamma\right](\mathbf{p^w} - \mathbf{mc^w}) + \Lambda^{-1}\mathcal{P}^{-1}\boldsymbol{\sigma}$$

with $\Lambda, \Gamma, \boldsymbol{\sigma}$ evaluated at $\mathbf{p^r}(\mathbf{p^w})$. Higher retail pass-through (bigger $\mathcal{P}$) shrinks the wholesale markup: the wholesaler's own price moves retail prices more than one-for-one.

- General (non-diagonal) $\mathcal{P}$: $\mathcal{P}'\Lambda$ is no longer diagonal, so the clean split fails. Split instead on $D = \text{diag}(\Omega_w)$, iterate $\mathbf{p^w} - \mathbf{mc^w} \leftarrow\!\shortmid D^{-1}(D - \Omega_w)(\mathbf{p^w} - \mathbf{mc^w}) + D^{-1}\boldsymbol{\sigma}$ with no convergence guarantee, or use Newton with the pass-through Jacobian.

What's the point? Vertical settings

Why do we care about pass-through in vertical settings?

- The wholesaler's effective demand derivative is $\mathcal{P}'\Delta_r$: with low retail pass-through the manufacturer faces a flatter response and takes a bigger margin; with high pass-through the retailer is doing the marking up.
- When retail pass-through is high we think that downstream firms have significant market power.
- This also means the scope for elimination of double marginalization is large.
  - Higher pass-through should be associated with more efficient vertical mergers? (Not sure this has been tested)
- A huge class of models (post-and-hold, minimum markups, RPM, vertical contracts in the second IO course) adds an extra parameter between wholesale and retail prices. All of them are a statement about $\mathcal{P}$.

Taxes and the data

Specific vs. Ad Valorem Taxes: The ζ Map Already Told Us

Producer price $\mathbf{p}$, consumer price $\mathbf{p^c}$. A specific tax t adds to cost; an ad valorem tax τ scales the consumer price, $\mathbf{p^c} = (1+\tau)\mathbf{p}$, i.e. $\mathcal{P} = (1+\tau)\mathbb{I}_J$ on the previous slide:

$$\text{specific:} \quad \mathbf{p} - \mathbf{mc} - t\mathbf{1} = \Lambda^{-1}[\mathcal{H} \odot \Gamma](\mathbf{p} - \mathbf{mc} - t\mathbf{1}) + \Lambda^{-1}\boldsymbol{\sigma}$$

$$\text{ad valorem:} \quad \mathbf{p} - \mathbf{mc} = \Lambda^{-1}[\mathcal{H} \odot \Gamma](\mathbf{p} - \mathbf{mc}) + \frac{1}{1+\tau}\Lambda^{-1}\boldsymbol{\sigma}$$

- A specific tax leaves the markup equation alone: the firm re-optimizes against a higher cost and pass-through is ρ from before.
- An ad valorem tax taxes the markup: the share term is scaled by $1/(1+\tau)$, so the producer gives up markup before any demand response. For the same revenue, ad valorem gives lower consumer prices under market power (Delipalla Keen 1992; Anderson, de Palma, Kreider 2001 for logit oligopoly).
- Single-product logit, rivals fixed: specific tax $\rho = 1 - \sigma_j$; ad valorem consumer-price pass-through is smaller still.
- US excise taxes (gasoline, cigarettes, alcohol, the Seattle soda tax) are specific; sales taxes are ad valorem. Conlon Rao (2025) compare volume, sales and ethanol taxes on spirits this way.

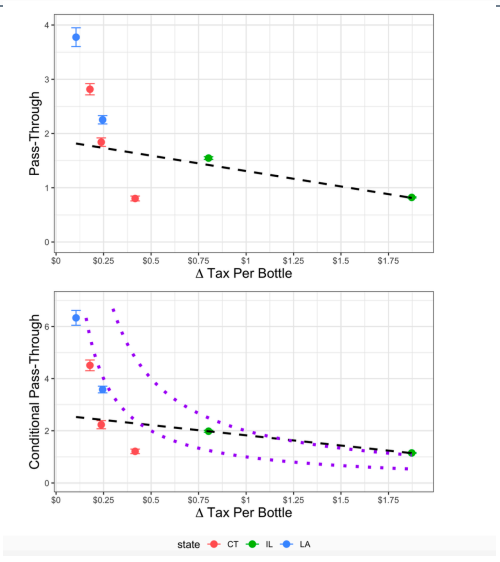
A simple empirical exercise (Conlon Rao, AEJ:EP 2020)

- US states tax distilled spirits by volume (a specific tax)
- We see tax changes in three states (Connecticut, Louisiana, Illinois)
- Three popular bottle sizes (750mL, 1L, 1.75L)
- Run a regression at a 3 month difference product by product

$$\Delta p_{jst} = \rho_{jst}(\mathbf{X}, \Delta\tau) \cdot \Delta\tau_{jt} + \beta \Delta x_{jst} + \gamma_j + \gamma_t + \epsilon_{jst}$$

- Only innovation: allow $\rho_{jst}(\mathbf{X}, \Delta\tau)$ to vary by size of tax increase.
- Prediction from the theory so far: something below one for a logit-like product, around one with constant markups, above one only with very convex demand.

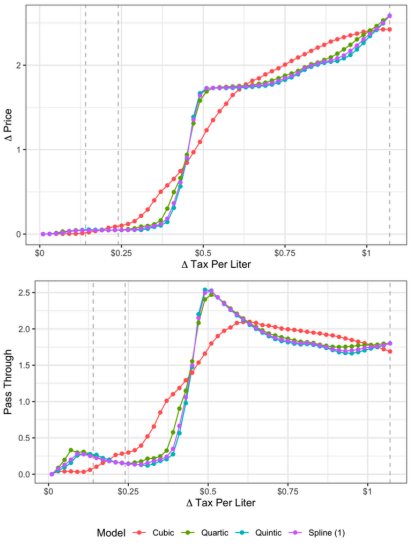
Goes horribly wrong



- Regression coefficients are huge > 1 and all over the map
- Conditional on a price change even larger > 3
- Seem to coincide with the purple line

Is PT a structural parameter?

- Price points are a big problem here
- Estimated an ordered probit with flexible $\beta(\Delta\tau)$
- Pass-through depends on where you are; not stable at all (!)
- Maybe structural interpretations of pass-through parameter are a bad idea!
- The theory is about a smooth FOC. Retail prices move in discrete steps ($X.99), so a 30 cent tax either does nothing or moves the price a dollar. Averaged over products that can look like anything.



Is It Chain Pricing? DellaVigna Gentzkow (QJE 2019), Butters Sacks Seo (AER 2022)

- DellaVigna and Gentzkow (2019): US retail chains set nearly uniform prices across their stores despite large differences in local demand and income. So a natural excuse for strange local tax pass-through: the chain prices nationally, the local tax is absorbed or spread across all stores.
- Butters, Sacks, Seo (2022) test it directly: 68 local excise tax changes (beer, liquor, cigarettes, soda) and 76 sales tax changes, 35,151 stores in 96 multi-state chains.
  - Local cost shocks are passed through to local prices, at similar rates for national and local chains, and about one-for-one in levels (Sangani's benchmark).
  - No spillovers to unaffected stores in the same chain.
- So chains price uniformly with respect to demand but respond locally to costs. Uniform pricing is not the explanation for the spirits results; price points and the discreteness of retail pricing are.
- Practical upshots: (a) local tax changes are clean cost shifters for the supply side of a demand model (the Conlon Rao supply moments); (b) the elasticity you back out from a tax change is the elasticity of the stores that actually changed price, with the price-point weighting from the previous slide.

Some challenges

- Pless and van Benthem (AEJ:Applied 2019): use pass-through > 1 as a test for market power? Only if you are willing to rule out convex demand under competition, and to assume constant marginal cost. The converse fails too: $\rho = 1$ in levels is not evidence of competition. Shift-invariant demand delivers it at any markup (Sangani 2026).
- Chetty, Looney, Kroft (AER 2009)

$$\log q\left(p, \tau^S\right) = \alpha + \beta \log p + \theta_\tau \beta \log\left(1 + \tau^S\right)$$
$$\theta_\tau = \frac{\partial \log q}{\partial \log\left(1 + \tau^S\right)} \Big/ \frac{\partial \log q}{\partial \log p} = \frac{\varepsilon_{q,1+\tau^s}}{\varepsilon_{q,p}}$$

- Use θ as evidence of "tax salience" (is it?). With imperfect competition the posted price responds to the tax too, so the incidence and excess-burden formulas change: Kroft, Laliberté, Leal-Vizcaíno, Notowidigdo (ReStud 2024) derive them with salience and market power together, find near-complete pass-through of sales taxes, consumers who respond more to prices than to taxes, and a marginal excess burden larger than the standard formulas give.
- Can we identify markups from pass-through alone...?!? Only jointly with curvature. Which is where we started.